# Curriculum Vitae

**Tirtharaj Dash**
Assistant Professor (Gr-II) & Doctoral Researcher

**Office:** D-168
Department of Computer Sc. and Information Systems
Birla Institute of Technology and Science (BITS) Pilani
K.K. Birla Goa Campus, Goa 403726, INDIA

## Personal Details

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|---|---|---|---|
| **D.O.B.** | July 03, 1991 | **E-mail(O)** | *tirtharaj@goa.bits-pilani.ac.in* |
| **Languages** | English, Hindi, Odia | **E-mail(P)** | *dashtirtharaj@gmail.com* |
| **Father's Name** | Mr. Lalitmohan Dash | **WWW** | https://tirtharajdash.github.io |
| **Mother's Name** | Mrs. Bijayanti Dash | **Tel(O)** | (+91) 832 2580 322 |
| **Residence** | Qtr. No. D-60<br>BITS Pilani, K.K. Birla Goa Campus<br>Zuari Nagar<br>Goa 403 726, INDIA | **Mobile(P)** | (+91) 8698567190 |

## Education

**Birla Institute of Technology and Science Pilani, India**

**Doctor of Philosophy (Ph.D.)**, Computer Science, **December 2016 – ongoing**
*Topic:* Search as a Game in Machine Learning
*Research areas:* Stochastic Search, Machine Learning, Inductive Logic Programming
*Advisor:* Prof. Ashwin Srinivasan

**Veer Surendra Sai University of Technology (formerly UCE), Burla, India**

**Master of Technology (M.Tech.)**, Computer Sc. & Engg., May **2014**, GPA: **9.78/10**
*Dissertation:* Pattern Recognition using Soft Computing Approaches
*Research areas:* Machine Learning, Pattern Recognition
*Advisor:* Prof. H. S. Behera

**National Institute of Science and Technology, Berhampur, India**

**Bachelor of Technology (B.Tech.)**, Information Technology, August **2012**, GPA: **8.91/10**
*Research areas:* Machine Learning, Pattern Recognition, Parallel Computing
*Advisors:* Prof. Motahar Reza, Prof. Subhagata Chattopadhyay

## Major Awards, Achievements & Scholarships

- **AWSAR Award 2019** from DST, Govt. of India (Country Rank: **9** in Best-100 Category) [Read]
- **Best Student Research Paper Award** from Springer's Machine Learning Journal in ILP-2018 (USD 750)
- **Travel Grant** (800 EURO) from EurAI to attend ACAI-2018, Ferrara, Italy during August 27-31, 2018
- **Summer Research Fellowship-2016** from NIST Berhampur (May-July, 2016)
- **IASc-INSA-NASI Fellowship-2015** to work with Prof. Nikhil R. Pal, ISI Kolkata (May-July 2015)
- **GATE-2015** (IIT-K; Score: 413), **GATE-2012** (IIT-D, Score: 407) (*Qualified*)
- **UGC NET-2014**, Organized by UGC New Delhi, Score: 212; 60.57% (*Qualified*)
- **University Silver Medal** for Best Post Graduate in CSE, Batch: 2012-2014 at VSSUT Burla
- **Institute Silver Medal** for Best Graduate in IT, Batch: 2008-2012 at NIST Berhampur
- GATE Scholarship from MHRD, India during M.Tech (2012-2014)

## Research Interests

Deep Learning, Machine Learning, Inductive Logic Programming, Optimization

## Research Experiences/Fellowships

**Summer Research Fellowship-2016** May-July, 2016
**Supervisor:** Dr. Prabhat K. Sahu, **Center for Multiscale Modeling, NIST Berhampur**
**Project Title:** Protein Sequence Analysis using Machine Learning Methods
**Details:** Study of application of different ML approaches such as Neural Nets, Probability Theory and Statistics to analyze various protein sequences.

**IASc-INSA-NASI Summer Research Fellowship-2015** May-July 2015
**Supervisor:** Prof. (Dr.) Nikhil R. Pal, *ECSU*, **Indian Statistical Institute (ISI) Kolkata**
**Project Title:** Studying the effects of various distance measures on Self Organizing Feature Map
**Details:** The standard Self Organizing Feature Map (SOFM) works with Euclidean distance measure. The objective was to study the effects of non-Euclidean distance measures such as 1-norm distance, Mahalanobis distance, Chebyshev/Infinity-norm distance, $p$-norm ($p$ is even) distance on different properties of SOFM. Major properties includes training performance, topology preservation and density matching.

## Publications in Journals (SCI/SCI-E/SCOPUS)

12. R.R. Swain, **T. Dash**, P.M. Khilar, "A Lightweight Approach to Automated Fault Diagnosis in Wireless Sensor Networks", *IET Networks*, 2020; DOI: 10.1049/iet-net.2019.0117; Publisher: IET. [Emerging-SCI]
11. R.R. Swain, **T. Dash**, P.M. Khilar, "A Complete Diagnosis of Faulty Sensor Modules in a Wireless Sensor Network", *Ad Hoc Networks*, vol. 93, 2019; DOI: 10.1016/j.adhoc.2019.101924; Publisher: Elsevier. [SCI-E]
10. **T. Dash**, S.N. Dambekodi, P.N. Reddy, A. Abraham, "Adversarial Neural Networks for Playing Hide-and-Search Board Game Scotland Yard", *Neural Computing and Applications*, 2018; DOI: 10.1007/s00521-018-3701-0; Publisher: Springer. [SCI-E]
9. **T. Dash**, H.S. Behera, "A Comprehensive Study on Evolutionary Algorithm based Multilayer Perceptron for Real-world Data Classification under Uncertainty", *Expert Systems*, 2018; DOI: 10.1111/exsy.12327; Publisher: John Wiley & Sons. [SCI-E]
8. R.R. Swain, P.M. Khilar, **T. Dash**, "Fault Diagnosis and its Prediction in Wireless Sensor Networks using Regressional Learning to achieve Fault Tolerance", International Journal of Communication Systems, 2018; DOI: 10.1002/dac.3769; Publisher: John Wiley & Sons. [SCI-E]
7. R.R. Swain, P.M. Khilar, **T. Dash**, "Multi-Fault Diagnosis in WSN using Hybrid meta-heuristic trained Neural network," *Digital Communications and Networks*, 2018; DOI: 10.1016/j.dcan.2018.02.001; Publisher: Elsevier. [SCI-E]
6. R.R. Swain, P.M. Khilar, **T. Dash**, "Neural network based automated detection of link failures in wireless sensor networks and extension to a study on the detection of disjoint nodes," *Journal of Ambient Intelligence and Humanized Computing*, 2018; DOI: 10.1007/s12652-018-0709-3; Publisher: Springer. [SCI-E]
5. P.P. Pai, **T. Dash**, S. Mondal, "Sequence-based Discrimination of Protein-RNA Interacting Residues using a Probabilistic Approach," *Journal of Theoretical Biology*, vol. 418, pp. 77-83, 2017; DOI: 10.1016/j.jtbi.2017.01.040; Publisher: Elsevier. [SCI]
4. R.R. Swain, **T. Dash**, P.M. Khilar, "An Effective Graph-theoretic approach towards Simultaneous Detection of Fault(s) and Cut(s) in Wireless Sensor Networks," *International Journal of Communication Systems*, 30(13), 2017; DOI 10.1002/dac.3273; Publisher: John Wiley & Sons. [SCI-E]
3. **T. Dash**, "A study on intrusion detection using neural networks trained with evolutionary algorithms," *Soft Computing*, 21(10), 2687-2700 (2017); (Online first: 2015), DOI: 10.1007/s00500-015-1967-z; Publisher: Springer. [SCI-E]
2. **T. Dash**, "Automatic Navigation of Wall Following Mobile Robot using Adaptive Resonance Theory of Type-1," *Biologically Inspired Cognitive Architectures*, 12, 2015, pp. 1-8; DOI: 10.1016/j.bica.2015.04.008; Publisher: Elsevier. [SCI-E]
1. **T. Dash**, P.K. Sahu, "Gradient Gravitational Search: An Efficient Metaheuristic Algorithm for Global Optimization," *Journal of Computational Chemistry*, 36(14), 2015, pp. 1060-1068; DOI: 10.1002/jcc.23891; Publisher: John Wiley & Sons. [SCI]

## Publications in Conference Proceedings

14. S. Yalburgi, **T. Dash**, R. Hebbalaguppe, S. Hegde, A. Srinivasan, "An Empirical Study of Iterative Knowledge Distillation for Neural Network Compression", In proc: *28th European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning (ESANN 2020)*, April 22-24, 2020, Bruges, Belgium. [CORE: B]
13. **T. Dash**, A. Srinivasan, R.S. Joshi, A. Baskar, "Discrete Stochastic Search and its Application to Feature-Selection for Deep Relational Machines", In proc: *28th International Conference on Artificial Neural Networks (ICANN 2019)*, September 17-19, 2019, Munich, Germany; DOI: 10.1007/978-3-030-30484-3_3. [CORE: B]
12. **T. Dash**, A. Srinivasan, L. Vig, O.I. Orhobor, R. D. King, "Large-Scale Assessment of Deep Relational Machines", In proc: *28th International Conference on Inductive Logic Programming (ILP 2018)*, September 2-4, 2018, Ferrara, Italy; DOI: 10.1007/978-3-319-99960-9_2. (Winner of Best Student Paper Award) [CORE: B]
11. P.S.M. Saladi, **T. Dash**, "Genetic Algorithm based Oversampling Technique to Learn from Imbalanced Data", *In proc: 7th International Conference on Soft Computing for Problem Solving (SocProS 2017)*, December 23-24, 2017, IIT Bhubaneswar; DOI: 10.1007/978-981-13-1592-3_30.
10. S. Iyer, S. Chaturvedi, **T. Dash**, "Image captioning based image search engine: an alternative to retrieval by metadata", *In proc: 7th International Conference on Soft Computing for Problem Solving (SocProS 2017)*, December 23-24, 2017, IIT Bhubaneswar; DOI: 10.1007/978-981-13-1595-4_14.
9. R.R. Swain, **T. Dash**, P.M. Khilar, "Investigation of RBF kernelized ANFIS for Fault Diagnosis in Wireless Sensor Networks", *In proc: International Conference on Computational Intelligence (ICCI 2017)*, IIT Kanpur, India (December 6-8, 2017); DOI: 10.1007/978-981-13-1135-2_20.
8. A. Saboo, A. Sharma, **T. Dash**, "GASOM: Genetic Algorithm assisted Architecture Learning in Self Organizing Maps", in proc: *24th International Conference on Neural Information Processing (ICONIP 2017)*, Guangzhou, China (November 14-18, 2017), **LNCS** vol. 10634; DOI: 10.1007/978-3-319-70087-8_25. [CORE: A]
7. P.N. Reddy, S.N. Dambekodi, **T. Dash**, "Towards Continuous Monitoring of Environment under Uncertainty: A Fuzzy Granular Decision Tree Approach", *In proc: Development aspects of Intelligent Adaptive Systems (DIAS); Innovations in Software Engineering Conference (ISEC-2017)*, Jaipur, India (Feb 5-7, 2017).
6. T. Nayak, **T. Dash**, D.C. Rao, P.K. Sahu, "Evolutionary Neural Networks versus Adaptive Resonance Theory Net for Breast Cancer Diagnosis," *In Proc: International Conference on Informatics and Analytics* (*ICIA'16*), Pondicherry, India (August 25-26, 2016); DOI: 10.1145/2980258.2980458, Publisher: **ACM**.
5. **T. Dash**, T. Nayak, R.R. Swain, "Controlling Wall Following Robot Navigation Based on Gravitational Search and Feed Forward Neural Network," *In proc: 2nd International Conference on Perception and Machine Intelligence* (*PerMIn'15*), C-DAC, Kolkata, India, 2015; pp. 196-200; DOI: 10.1145/2708463.2709070, Publisher: **ACM** [SIGCHI, SIGGRAPH].
4. **T. Dash**, S.K. Nayak, H.S. Behera, "Hybrid Gravitational Search and Particle Swarm Based Fuzzy MLP for Medical Data Classification," *In proc: International Conference on Computational Intelligence in Data Mining* (*ICCIDM-2014*), Burla, India (December 20-21, 2014), **Smart Innovation, Systems and Technologies**, Volume 31, 2014, pp. 35-43; DOI: 10.1007/978-81-322-2205-7_4, **Springer**.
3. J. Nayak, N. Sahoo, J.R. Swain, **T. Dash**, H.S. Behera, "GA based Polynomial Neural Network for Data Classification," *In proc: 13th International Conference on Information Technology* (*ICIT-2014*), Bhubaneswar, India (December 22-24, 2014), DOI: 10.1109/ICIT.2014.55, pp. 234-239, Publisher: **IEEE**.
2. **T. Dash**, S.R. Sahu, T. Nayak, G. Mishra, "Neural Network Approach to Control Wall-Following Robot Navigation," *In proc: IEEE International Conference on Advanced Communication, Control and Computing Technologies* (*ICACCCT-2014*), Tamilnadu, India (May 8-10, 2014), pp. 1072-1076, ISBN No. 978-1-4799-3914-5/14, Publisher: **IEEE**.
1. R. Mohanty, **T. Dash**, B. Khan, S.P. Dash, "An Experimental Study of a Novel Move-to-Front-or-Middle (MFM) List Update Algorithm," *In proc: First International Conference on Applied Algorithms* (*ICAA-2014*), Kolkata, India (January 13-15, 2014), **LNCS** vol. 8321, 2014, pp. 187-197; DOI: 10.1007/978-3-319-04126-1_16), **Springer International Publishing**.

## Research Groups and Memberships

- ACM Professional Membership (No. 6108003)
- Regular Member, Asia Pacific Neural Network Society (No. 17021031)
- Data Science Research Group, BITS Pilani, Goa
- Annotate Biomolecules Computationally (ABC) Group, BITS Pilani, Goa

- Regular Member of Machine Intelligence Research Labs, USA

## Academic Experiences

**Birla Institute of Technology and Science Pilani (BITS Pilani)** August 2015 – till date
*Lecturer, Data Science Research Group, Dept. of Computer Science, BITS Pilani, K.K. Birla Goa Campus, India*
*Responsibilities:* Teaching and Research, Co-curricular activities (seminar, conferences, talks, website), Coach for ACM-ICPC, mentoring BE and ME theses, and various student projects.
*In-charge of courses:* Neural Networks (Deep Learning), Machine Learning
*Courses teaching/taught:* Neural Networks (Deep Learning), Machine Learning, Artificial Intelligence, Computer Programming, Database Systems

**National Institute of Science and Technology (NIST) Berhampur, India** June 2014 – August 2015
*Assistant Professor, School of Comp. Sc.:* Involved in teaching and research activities in the institute.
*Courses taught:* Analysis and Design of Algorithms, Computer and Object Oriented Programming

**Veer Surendra Sai University of Technology Burla, India** August 2012 – May 2014
*Teaching Assistant as per MHRD during M.Tech.:* Conducted labs for UG students.

## Academic Services

**Peer-review for Journals:** Robotica (Cambridge Press) • Neural Processing Letters (Springer) • Information Fusion (Elsevier) • Computers & Security (Elsevier) • Paladyn Journal of Behavioral Robotics • Ain Shams Engineering Journal (Elsevier) • Proceedings of the National Academy of Sciences India • Computer Methods and Programs in Biomedicine (Elsevier) • Journal of Information Processing Systems • Wireless Personal Communications (Springer) • International Journal of Image Mining (Inderscience) • International Journal of Image and Data Fusion (Taylor & Francis) • International Journal of Communication Systems (Wiley) • IEEE Access • IEEE Systems Journal • IEEE Transactions on Industrial Electronics • Cognitive Systems Research (Elsevier) • IEEE Transactions on Cybernetics • Data Technologies and Applications (Emerald)

**Peer-review for Conferences/Workshops:** IJCNN-2020 (Glasgow, UK) • ICANN-2019 (Germany) • IJCNN-2019 (Hungary) • ILP-2018 (Italy) • AIAPP 2018 (Dubai, UAE) • ICCI-2017 (IIT-K, India) • ACM ICIA'16 (India) • $20^{th}$ WMSCI-2016 (USA) • IEEE WCI-2016 (IIT Kanpur, India) • FICTA-2015 (India) • CITSA-2014 (USA)

**Peer-review for Book chapters:** Robot Operating System (ROS), Springer • Emergent Research on the Application of Optimization Algorithms, IGI Global

**Conference associations:** Technical Committee Member, IJCNN 2020 • Reviewer, ILP 2019 • Programme Committee Member, ICANN 2019 • Technical Committee Member, IJCNN 2019 • Organizing Committee Member, ICCI-2017, IIT-K, India • ISEC-2016 • ICCIDM-2014

**Competitions:** Coach for ACM ICPC

## Workshops/Schools Attended

4. Indo-German Spring School on Algorithms for Big Data, IIT Bombay, February 18-22, 2019 (also delivered a talk on: "Learning in Presence of Expert Knowledge: A talk on Inductive Logic Programming").
3. Summer School on Statistical Relational AI (ACAI), University of Ferrara, Italy, August 27-31, 2018.
2. WIPRO Mission10X Engineering Faculty Workshop, NIST Berhampur, December 2-4, 2014.
1. Intl. workshop on Cyber Physical Systems: Applications and Open Challeneges, IIT Hyd, March 22-23, 2013.

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